

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1668

Ansonia 345 kV
February 17, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1668. The Interconnection Customer proposes a 142.3 MW natural gas combined cycle facility interconnecting at the Ansonia 345 kV substation in the New Jersey area. The request was received on February 15, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 592 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$117,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ansona – Fanwood 138kV
Contingency: Locust Grove 345-138 kV Transformer Bank
Loading: 108.4% (Rating: 242 MVA)
DFAX: 5.8%
Pre-project loading: 105.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ansonia – Fanwood 138kV line	Reconductor line	\$458,000	26% (\$117,000)
	Total allocated to project		\$117,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.